$R_N = mg\cos\theta + \frac{mv^2}{r}\sin\theta$

$\frac{mv^2}{r}\cos\theta = mg\sin\theta + \mu R_N$

$\frac{mv^2}{r}\cos\theta = mg\sin\theta + \mu mg\cos\theta + \frac{\mu mv^2}{r}\sin\theta$

$\frac{v^2}{r} = g\tan\theta + \mu g + \frac{\mu v^2}{r}\tan\theta$

$\frac{v^2}{r}(1-\mu\tan\theta)$

$\frac{rg\tan\theta + \mu g}{1-\mu\tan\theta}$

JUNE 2025

REGIONAL MARITIME UNIVERSITY

DEPARTMENT: MARINE ENGINEERING

PROGRAMME: BME 27 LEVEL 200

COURSE: BME 202 MECHANICS OF MACHINES I

END-OF-SEMESTER TWO EXAMINATION

A

DATE	:	**JUNE, 2025**
TIME	:	**3 HOURS**
INSTRUCTIONS	:	**ANSWER THREE (3) QUESTIONS. QUESTIONS ONE AND TWO ARE COMPULSORY**

$\sqrt{\frac{rg(\tan\theta + \mu)}{1-\mu\tan\theta}}$

Q1. Explain any five of the following terms:

(i) **Mechanism** (ii) **Involute Gear** (iii) **Friction clutch**
(iv) **Kennedy's theorem** (v) **Degree of freedom** (vi) **Instantaneous center** [5 Marks]

Q2 The displacement function of the slider crank mechanism in fig Q2 is given as:

$$x = r\cos\theta + l - \frac{r^2}{2l}\left(\frac{1}{2} - \frac{1}{2}\cos 2\theta\right)$$ (All symbols have their usual meaning)

(i) If that the radius of the crank ***OA*** is 6cm and the connecting rod ***AB*** is 15cm long, find the velocity and acceleration of the piston ***B*** after time 5 seconds assuming that the crank is rotating anticlockwise at a constant speed of 150rev/min. [15marks]

$x = r\cos\theta + l - \frac{r^2}{2l}\left(\frac{1}{2} - \frac{1}{2}\cos 2\theta\right)$

$\dot{x} = -r\sin\theta\,\dot{\theta} - \frac{r^2}{2l}\left[\frac{1}{2}\sin 2\theta\,\dot{\theta}\right]$

$= -r\dot{\theta}\left[\sin\theta + \frac{r}{2l}\sin 2\theta\right]$

$= -r\dot{\theta}\left[\cos\theta\,\dot{\theta} + \frac{r}{2l}\cos 2\theta\,\dot{\theta}\right]$

$= -r\dot{\theta}^2\left[\cos\theta + \frac{r}{2l}\cos 2\theta\right]$

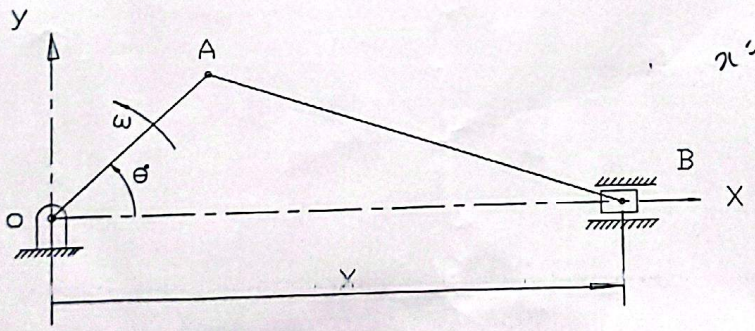

FIG Q2

Doc. No.: EXM/F/03	**Page No.:** 1/2	**Issue No.:** 1	**Last Update:** 15/04/08
Revision No.: 1	**Date:** 12/04/06	**Author:** EC	**Approved:** DAA

$\dot{x} = -(6\times10^{-2})(5\pi)\left[\sin(5\pi\times5) + \frac{6\times10^{-2}}{2\times15\times10^{-2}}\sin(2\times5\pi\times5)\right]$